In [14]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [15]:
```
data=pd.read_csv(r"C:\Users\abhis\Downloads\claimants (1).csv")
```

In [5]:
```
data.head()
```

Out[5]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>CASENUM</th> <th>ATTORNEY</th> <th>CLMSEX</th> <th>CLMINSUR</th> <th>SEATBELT</th> <th>CLMAGE</th> <th>LOSS</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

5	0	0.0	1.0	0.0	50.0	34.940
3	1	1.0	0.0	0.0	18.0	0.891
66	1	0.0	1.0	0.0	5.0	0.330
70	0	0.0	1.0	1.0	31.0	0.037
96	1	0.0	1.0	0.0	30.0	0.038

In [6]:
```
data.describe()
```

Out[6]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>CASENUM</th> <th>ATTORNEY</th> <th>CLMSEX</th> <th>CLMINSUR</th> <th>SEATBELT</th> <th>CLMAGE</th> <th>LOSS</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>

1340.000000	1340.000000	1328.000000	1299.000000	1292.000000	1151.000000	1340.000000
11202.001493	0.488806	0.558735	0.907621	0.017028	28.414422	3.806307
9512.750796	0.500061	0.496725	0.289671	0.129425	20.304451	10.636903
0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
4177.000000	0.000000	0.000000	1.000000	0.000000	9.000000	0.400000
8756.500000	0.000000	1.000000	1.000000	0.000000	30.000000	1.069500
15702.500000	1.000000	1.000000	1.000000	0.000000	43.000000	3.781500
34153.000000	1.000000	1.000000	1.000000	1.000000	95.000000	173.604000

In [7]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1340 entries, 0 to 1339
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   CASENUM   1340 non-null   int64  
 1   ATTORNEY  1340 non-null   int64  
 2   CLMSEX    1328 non-null   float64
 3   CLMINSUR  1299 non-null   float64
 4   SEATBELT  1292 non-null   float64
 5   CLMAGE    1151 non-null   float64
 6   LOSS      1340 non-null   float64
dtypes: float64(5), int64(2)
memory usage: 73.4 KB
```

In [8]:
```
data.shape
```

Out[8]:
```
(1340, 7)
```

In [9]:
```
data.nunique()
```

Out[9]:
```
CASENUM     1283
ATTORNEY       2
CLMSEX         2
CLMINSUR       2
SEATBELT       2
CLMAGE        68
LOSS         916
dtype: int64
```

In [16]:
```
data=data.dropna()
```

In [17]:
```
x = data[['CASENUM','LOSS','CLMSEX','CLMINSUR','SEATBELT','CLMAGE']]
y = from        sklearn.model_selection import  train_test_splitdata['ATTORNEY']
print(x.head())
print(y.head())
```

```
   CASENUM    LOSS  CLMSEX  CLMINSUR  SEATBELT  CLMAGE
0        5  34.940     0.0       1.0       0.0    50.0
1        3   0.891     1.0       0.0       0.0    18.0
2       66   0.330     0.0       1.0       0.0     5.0
3       70   0.037     0.0       1.0       1.0    31.0
4       96   0.038     0.0       1.0       0.0    30.0
0    0
1    1
2    1
3    0
4    1
Name: ATTORNEY, dtype: int64
```

In [18]:
```
from    sklearn.model_selection import  train_test_split
x_train,        x_test, y_train,        y_test  =       train_test_split(x,y,test_size  =       0.2,    random_
```

In [20]:
```
from    sklearn.linear_model    import  LogisticRegression
from    sklearn.metrics import  accuracy_score
from    sklearn.preprocessing   import  StandardScaler

scaler  =       StandardScaler()
x_train =       scaler.fit_transform(x_train)
x_test  =       scaler.transform(x_test)

model   =       LogisticRegression(max_iter=1000)
model.fit(x_train,      y_train)
y_pred  =       model.predict(x_test)
```

In [21]:
```
print("Accuracy:",      accuracy_score(y_test,  y_pred))
```

```
Accuracy: 0.6727272727272727
```

In [ ]: